

2022

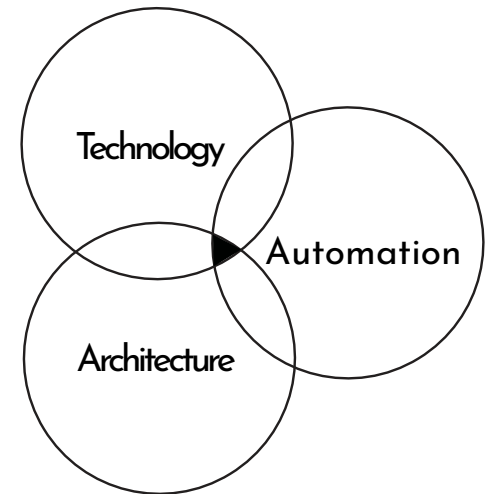
2025

POET FUGIO

Yenhsing Cheng

INDEX

	page	
<i># Past</i>	01	Orienteering Landscape / Reprogramming a far island and Reuse a WWII military spot
<i># Society</i>	08	1 House for All Housing / Transitional housing system during urban renewal
<i># Reality</i>	13	Banqiao Station Entrance Construction / System / Professional Work
<i># Crisis</i>	16	Earthen Ceramics / Rammed earth for post-wildfire community rebuilding and recovery
<i># Public</i>	21	SCI-Arc Gala Visual / AI-generated videos projected on the facade of DTLA Public Library



This architecture portfolio is based on my
selected works from 2022 to 2025



Regeneration of the Military Heritage



Orienteering

Instructor Shu-Chang Kung, Ling-Li Tseng
Duration 5 weeks (Summer 2022)
5-person Team Responsible for architectural design and programming

Site: Three Landscape Layers in a Former Military Spot

Located on Xiju Island in the Matsu Archipelago, this project occupies a far-far-away island at Taiwan's maritime edge.

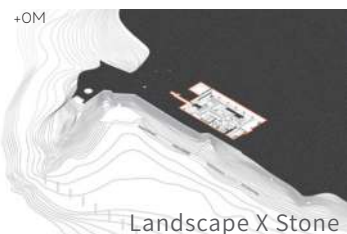
The project opens the former military site for public use. The site, Former Military Outpost No. 32 at the westernmost tip of Xiju, sits on a hilly terrain where sandstone cliffs meet the sea, shaped by Cold War frontline defense strategies. Its planting and spatial organization reflect a defensive logic, while the convergence of mountain and ocean defines a stark, strategic landscape condition.

Outpost No. 32 contains a rich and layered defense system organized by topography. The landscape on this island is a product of military necessity, shaped by soldiers stationed here 40-50 years ago. As a result, vegetation and ecological conditions vary according to elevation and military function.

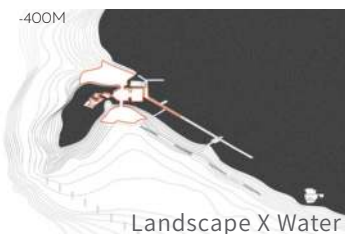
Based on elevation and existing infrastructure, three distinct landscape layers are identified:



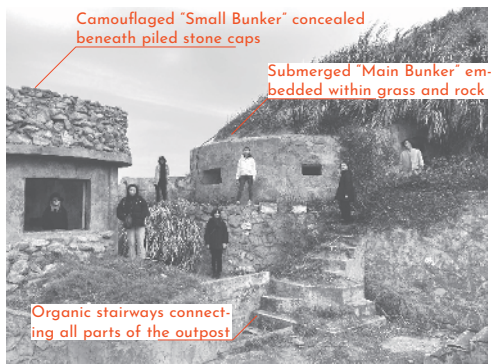
New planting strategies reinterpret Matsu's military landscape while restoring the rear hillside, re-covering it with newly formed terrain.



Stacked stone is used to define and differentiate functional spaces, creating interiors that feel as if they are carved directly from the mountain face.



Water is drained into the underground tunnel to form both public and private bathing spaces, while addressing persistent humidity issues within the subterranean structure.



Surface-level Bunkers embedded between earth mounds and cliff act as strategic nodes connecting different elevations—small bunkers serve as forward posts with over 270-degree defensive coverage and larger bunkers are carved into the hillside as rear defenses.

Semi-subterranean concrete living quarters are concealed within concrete structures, where narrow gaps introduce light to form administrative areas.

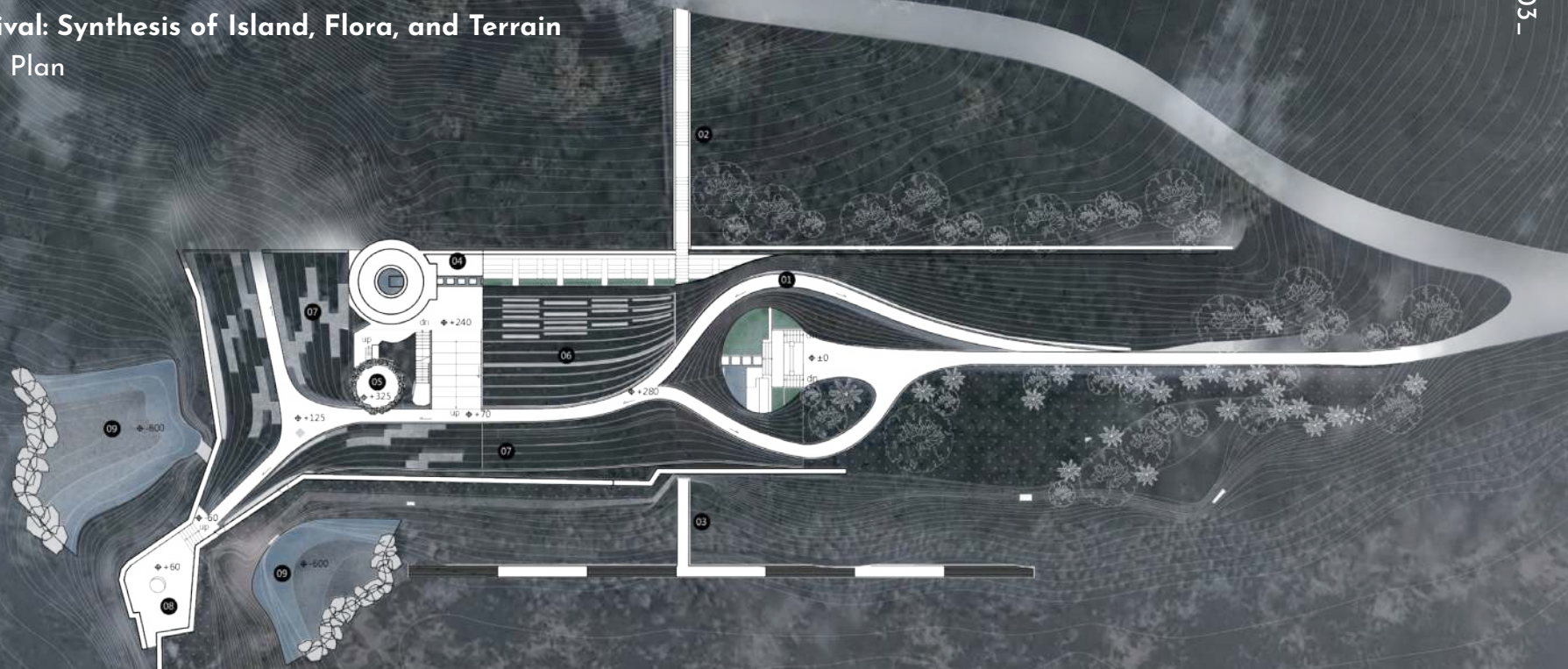
The underground tunnels extend closer to the coastline, positioning them nearer to potential threats, with chambers of varying scales distributed along both ends of the tunnel system.

+240M	
+200M	
+50M	
-400M	



Arrival: Synthesis of Island, Flora, and Terrain Site Plan

03_



The site is used by both residents and visitors, who move through it as part of daily routines rather than as a destination. Movement through the site defines the project more than fixed spaces.

Arrival Sequence	Slope Sequence	Arrival Sequence	Slope Sequence
01 Landscape Trail	03 Stepped Platform	08 Viewing Platform	02 Driveway
	04 Large Bunker	09 Bathing Area	
	05 Small Bunker		
	06 Viewing Platform		
	07 Rest Area		



Path: Axial flow from sea to mountain



RE: Programming

To reprogram the island as a brand, we see the undeveloped landscape of Xiju as an opportunity for people from urban environments to rediscover the senses they often overlook.

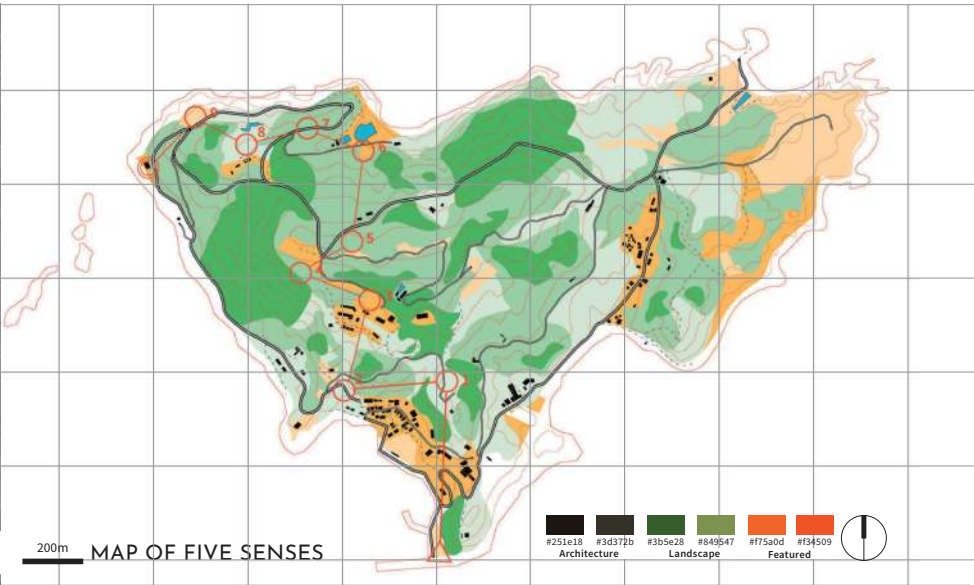
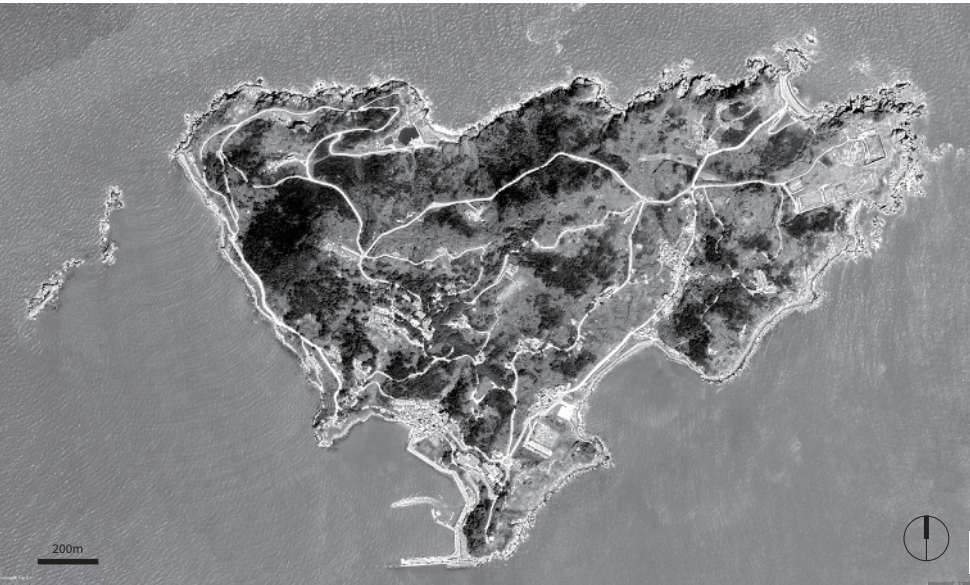
The *Orienteering Project* focuses on the five senses: sight, sound, smell, taste, and touch, as the ways the human body connects with the surrounding world.

By redesigning Military Site No. 32 and rethinking the orienteering system, we propose five different routes for visitors. Each route is designed around one of the senses, guiding people through a distinct experience of the landscape.

Through this approach, exploring Xiju Island becomes less about reaching a destination and more about experiencing the journey. Each visit can feel different, offering new moments of discovery every time.

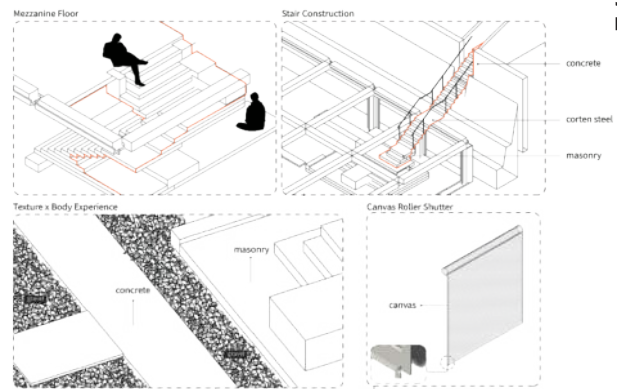


ORIENTEERING
ORIENTEERING
ORIENTEERING
ORIENTEERING
ORIENTEERING



Island Living Room
Transit Waystation
Coastal Bathhouse







This conceptual diagram is a collaborative work by the architectural team of this project.

1 House for All

Sustainable
Shared
Housing

Duration

8 months (2021F-2022S)

30-person Teamwork

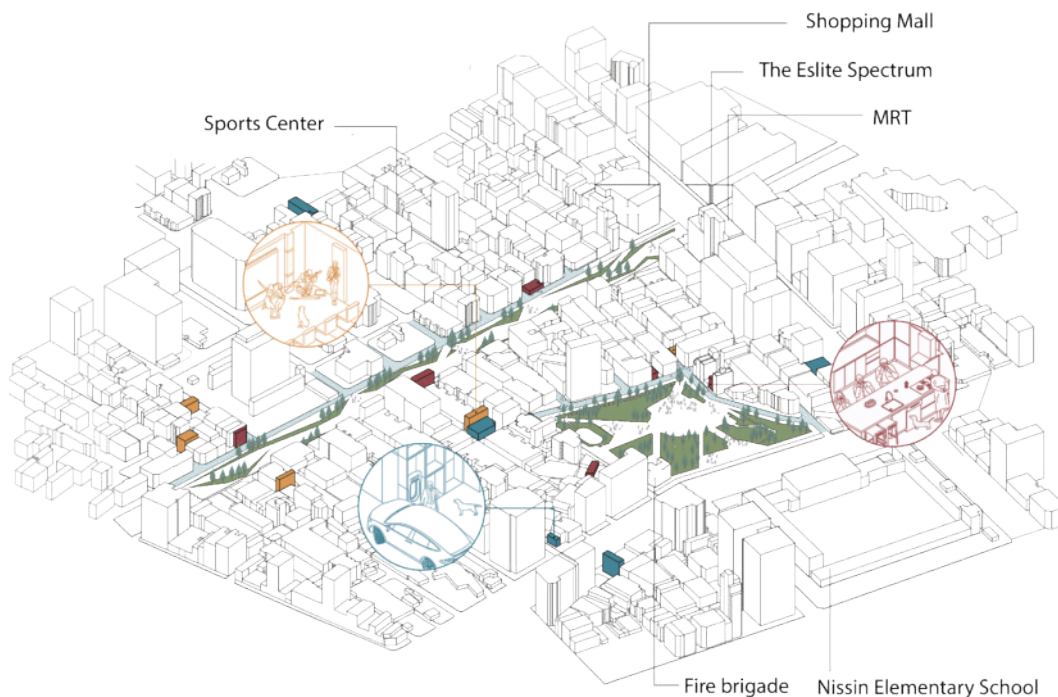
Responsible for design development and BIM coordination

This project is an international competition and the final work is a 1:1 house demonstration unit built up by ourselves.

What if housing could support communities through instability, instead of waiting for stability to return?

The average building lifespan in Taiwan is about 35-40 years. In Taipei, most housing are row houses or apartments built in the 1970s using reinforced concrete, which is no longer safe or sustainable. Although the government has encouraged citizens to rebuild through land integration, implementing urban renewal is difficult. The average renewal process takes 10 years. The ownership structure is complex, and many landowners have demolished their buildings while waiting. This has created many gaps in the neighborhood.

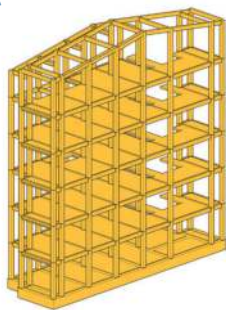
To combat inefficient urban renewal and energy overconsumption, we propose a new concept - building mid-way housing. This integrated sustainable solution can be replicated in different site conditions. This is not just a net-zero mid-way house, but also a social enterprise housing and an energy-efficient sharing hub for the city. By filling the gaps with these mid-way buildings, we can not only provide affordable housing but also add e-bike charging stations to the neighborhood, including some for e-cargo bikes to facilitate goods unloading.



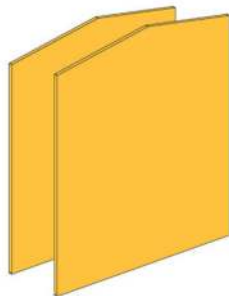
Building Life Cycle



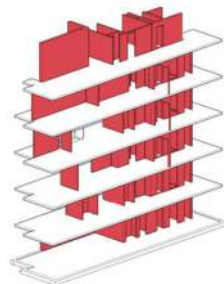
Foundation
100 years



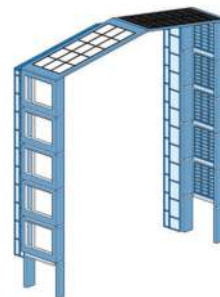
Structure
50-100 years



Insulated Wall
10-30 years



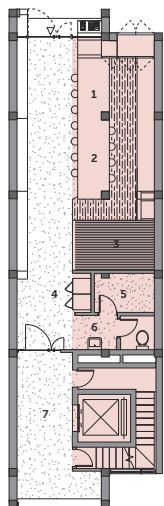
Partition
10-30 years



Facade
30-50 years

"1 House for All" is built with mass timber CLT and GLT. The building system is designed to be **fast constructed**, **completely disassembled**, and **easily reused in the next construction**. The project brings together elderly residents, families, and temporary occupants.

Ground_ Shared Space



Communities would get a place to gather, busy city dwellers could have the chance to communicate and fulfill their basic needs of living while rushing in and out. The shared space is a platform for people to find support in their daily lives. By providing these services and space, the income could support the house maintenance and help the business to provide more affordable housing to the neighborhood.



- 1 Dining Bar
- 2 Meeting Space
- 3 Reading Space
- 4 Self-served Laundromat
- 5 Storage Room
- 6 W/C
- 7 Circulation Core

2-5FL_ Dwelling Units

There are 4 floors of dwelling units in each housing. One floor is reserved for elder residents; the other floors are open to all generations. The design of the **party-core wall (Red Area)** helps efficiently integrate the pipe system and creates flexibility for the rest of the floor area for different use.

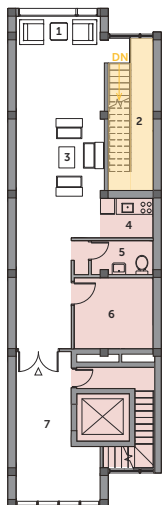


- 1 Bedroom
- 2 Kitchen
- 3 Dining Space
- 4 Living Room
- 5 Reading Room



- 6 W/C
- 7 Bathroom
- 8 Laundromat
- 9 Circulation Core

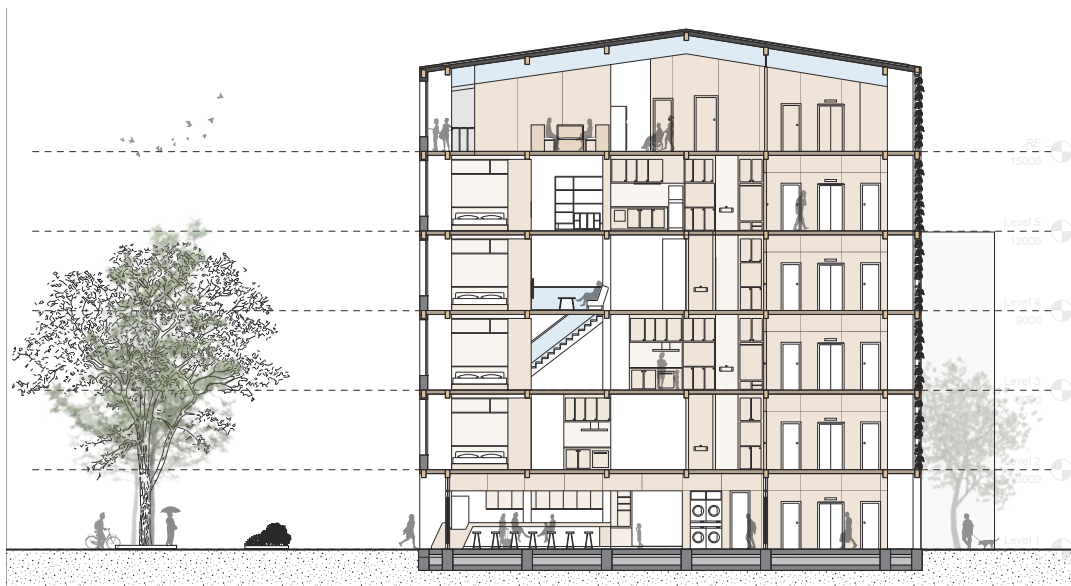
Roof_ Day Care Center



The **party-core wall (Red Area)** could provide enough storage space for the facility. The sunlight room could not only for activities but also as a thermal mass for the whole housing. When the temperature is cold outside, the ERV system would use the air here to provide fresh and warmer air.

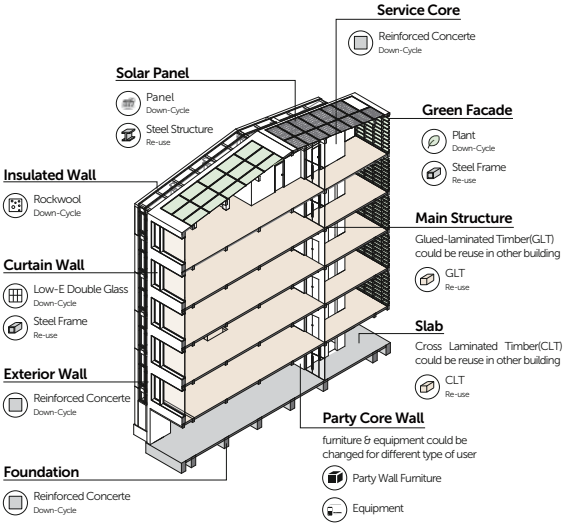


- 1 Social Corner
- 2 Sunlight Room
- 3 Dining Space
- 4 Shared Kitchen
- 5 W/C
- 6 Machine Room
- 7 Circulation Core

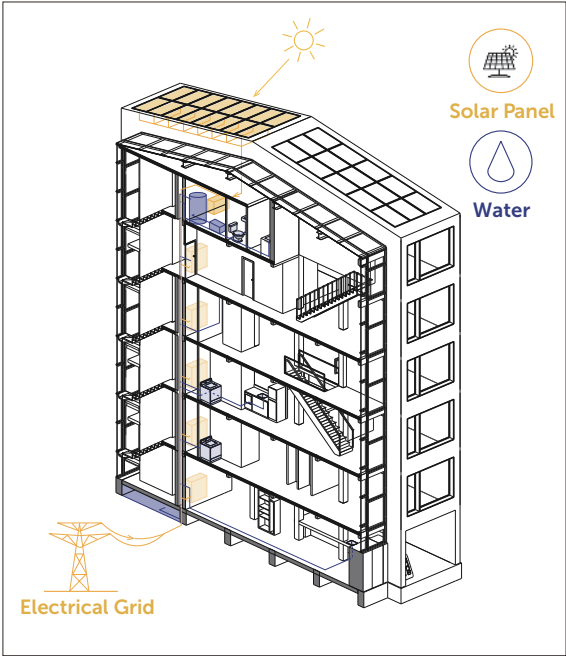


We are creating an electricity generating system specialized for a neighborhood. Local infrastructures can be powered, and the locals can benefit by acquiring electricity at lower costs. The operation mode is similar to an "OFF GRID" mode, the AC grid acts as a backup power source, and loads will be switched to AC input automatically while needed. We will use an Energy Management System (EMS) to control the charging period of our battery to lower the consumption peak.

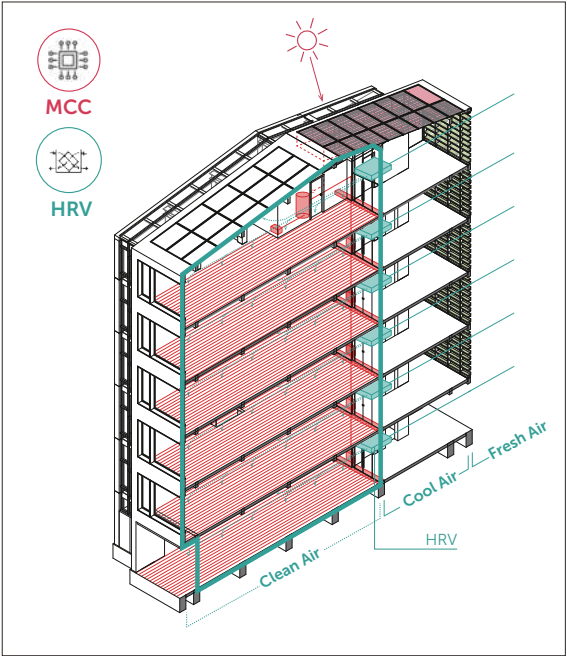
This building can save lots of energy compared to the traditional air conditioners, which consume most of the electrical energy in daily life. For radiation flooring, the solar water system can heat up the water in advance during the day and save it in the tank. After sunset, the indoor temperature can still be adjusted by the system. Through a switch on the pipe, the ERV system can use the heated air in the sunlight room when it's needed.



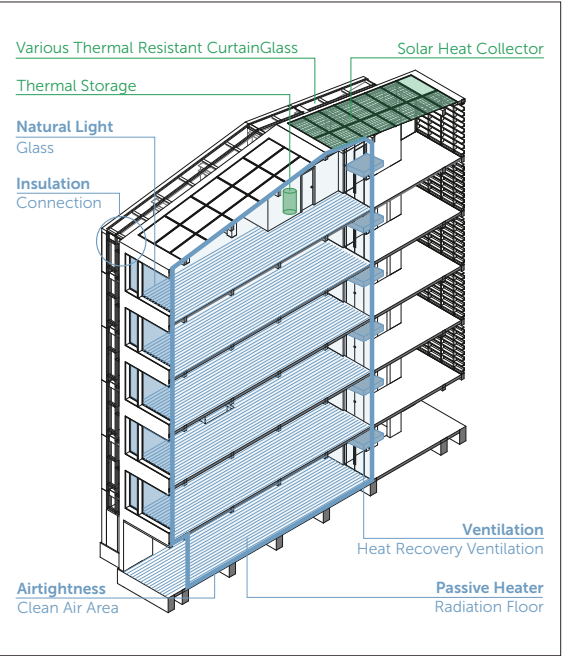
WATER & ELECTRICITY SYSTEM



ACTIVE CONTROL STRATEGY



PASSIVE CONTROL SYSTEM



1:1 ADU built by our team



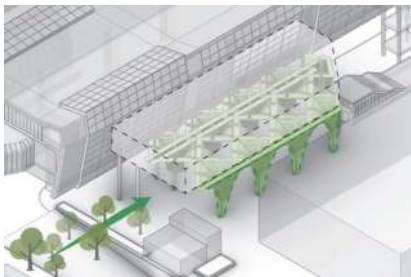


Professional Work, Square-Arch Architect & Associates

Banqiao Station Entrance

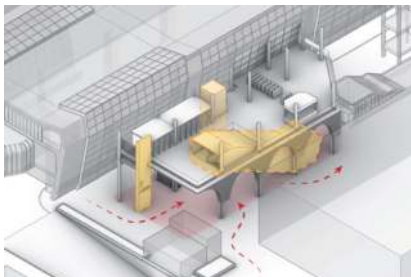
Area 885 sqm
Location Banqiao, New Taipei City, Taiwan
Budget NT\$29,920,000 (USD998,000)

We worked as a 3-person Team. Responsible for architectural design and coordination with outsourced team in India, where decisions often had to adjust to construction constraints



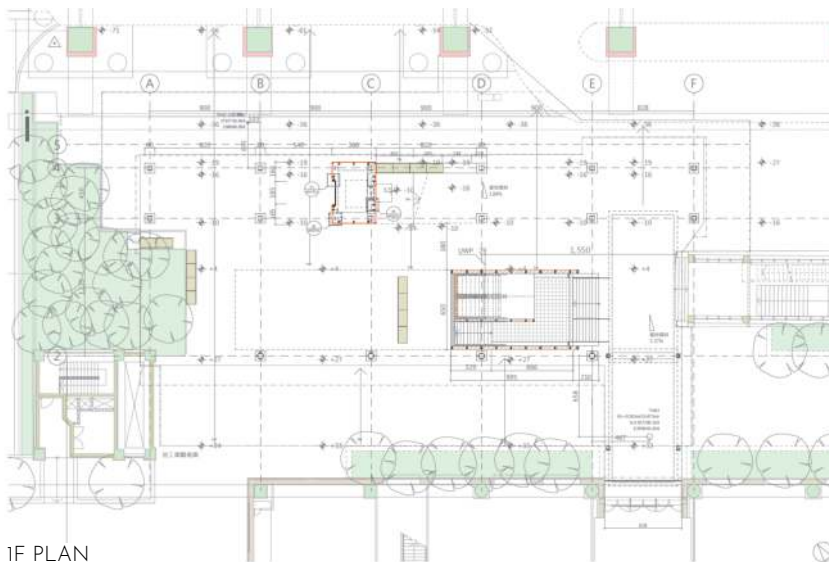
Extending the green belt

Landscape motifs are translated into arched structures, prolonging the natural atmosphere and forming a continuous urban green corridor.



Open and elevated ground plane

An open, double-height ground level enhances spatial permeability, while the loop line's signature color provides covered circulation and clearer entry points for daily commuters.



1F PLAN



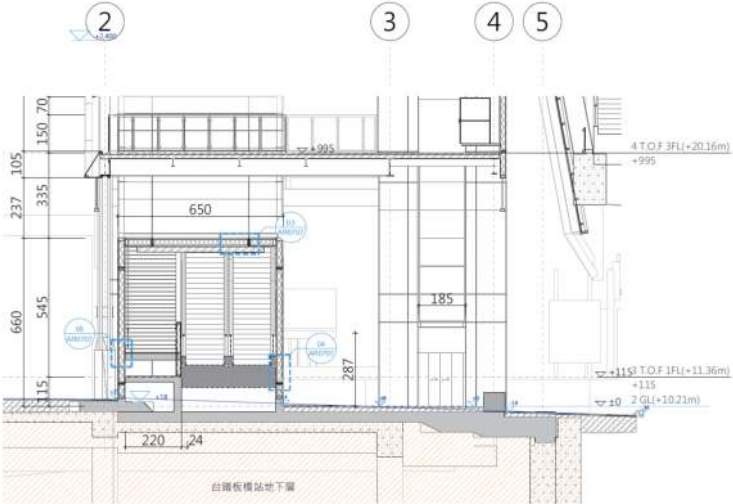
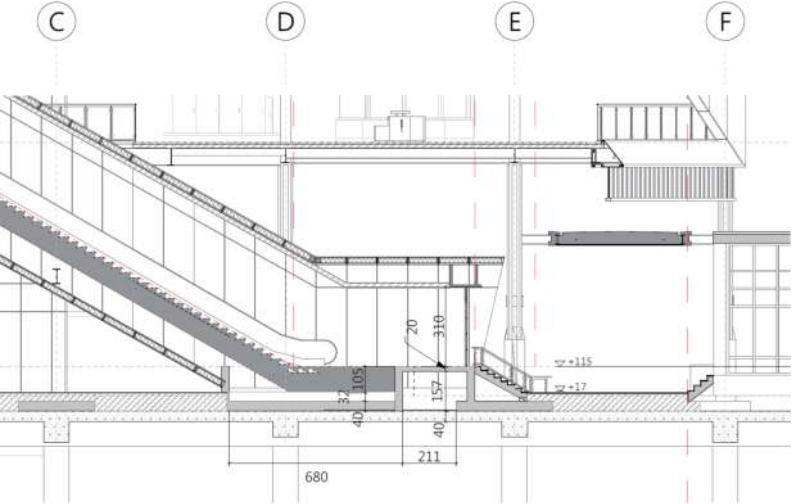
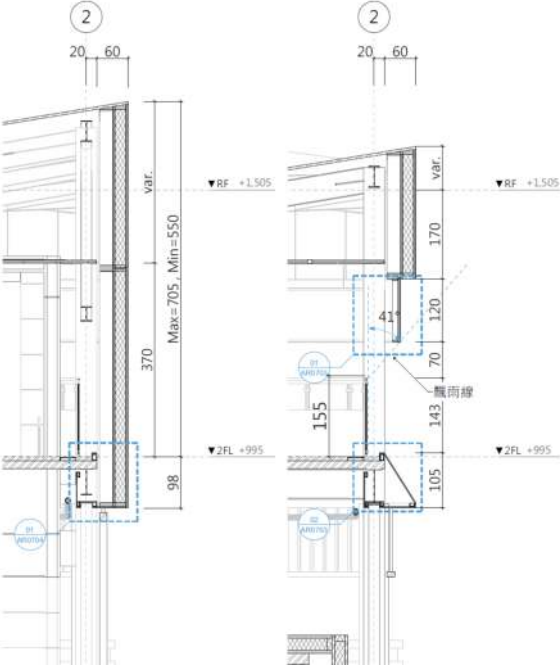
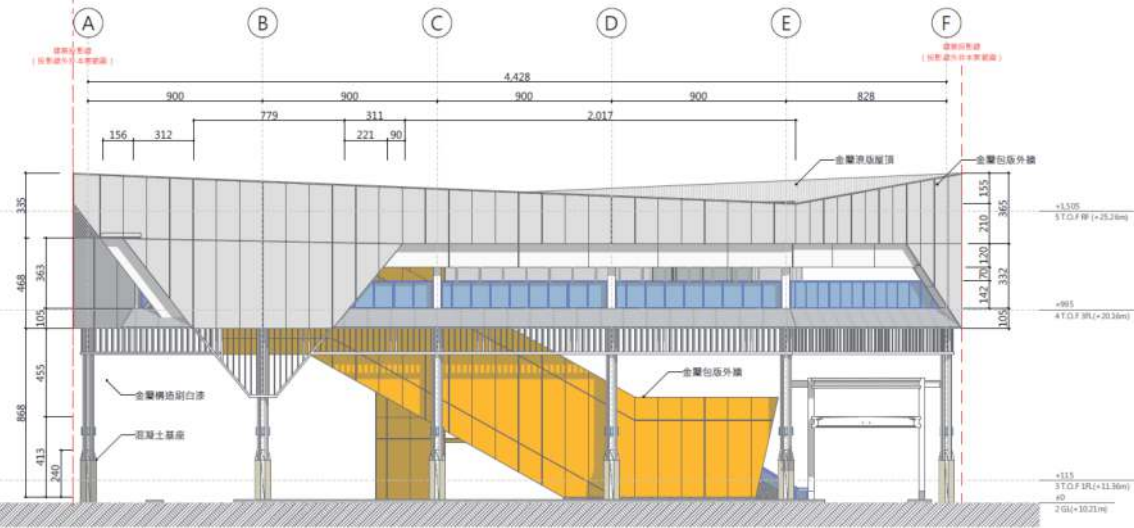
Inverted curtain wall system

Cantilevered 2m extension to enhance wind-driven rain shielding performance.

Rain-screen louvers

Integrated metal vertical fins and extended louver systems optimized for enhanced wind protection and coverage.







An ADU system supporting post-wildfire recovery in California

Earthen

Instructor

Herwig Baumgartner

Duration

15 weeks (Fall 2025)

7-person Teamwork

Responsible for structural design, prototype design, glazing and robotic 3D printing

After California Wildfires, how can architecture provide immediate shelter after disaster without becoming temporary in value?

Earthen was developed in response to the increasing frequency of wildfires in California and the prolonged displacement that follows. In the aftermath of disaster, housing is often treated as either temporary shelter or long-term reconstruction, leaving a gap where people are forced into unstable living conditions.

This project proposes a system of transitional ADUs that can be deployed quickly while maintaining spatial and material quality after wildfires. The units are designed for residents displaced by wildfires who need stable living conditions before permanent rebuilding. Using 3D-printed ceramics and rammed earth, the design explores construction methods that are both locally grounded and thermally resilient.

Sited at 3328 N Fair Oaks Avenue in Altadena, a lot left in ruin after the Eaton Fire, the project also introduces a community bathhouse paired with the housing units. Together, they form a small-scale recovery infrastructure that supports both daily life and collective healing.



The building envelope plays a central role. A robotically 3D-printed ceramic facade is designed to collect, channel, and retain rainwater. Material and fabrication are closely tied to the concept. Each ceramic module is 3D-printed in clay, then glazed and fired, resulting in an earth-toned palette that reflects the post-fire landscape. The surface texture draws from both erosion and water flow, linking destruction with regeneration.

Rather than treating temporary housing as disposable, Earthen frames it as part of a longer recovery process. The units are designed to remain useful over time, supporting a gradual return to stability instead of a sudden replacement. The project suggests that architecture can do more than provide shelter. The units are not meant to be discarded, but to remain in use as part of a longer recovery process and help rebuild a sense of community after disaster.



ZONE 1

Organic Extrusion

Directs rainwater down

ZONE 2

Water Channel Grooves

Carved to direct water flow

ZONE 3

Concealed Irrigation Pipe

Constant water infusion

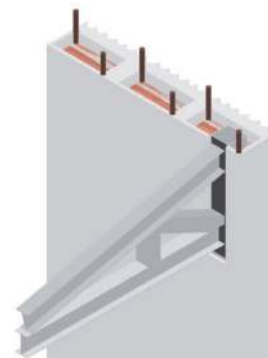
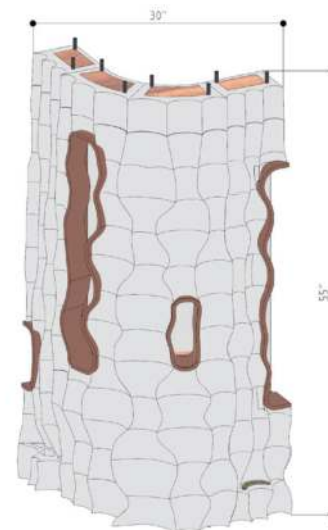
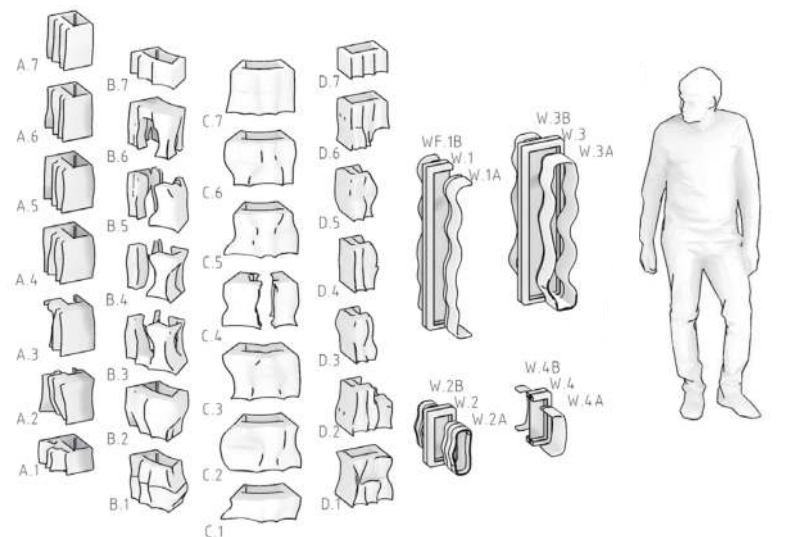
Larger Pockets

Indented cavities that collect water and supports small plants

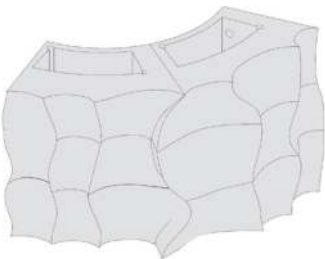
Drainage and Seepage

Allows access water to seep into the ground

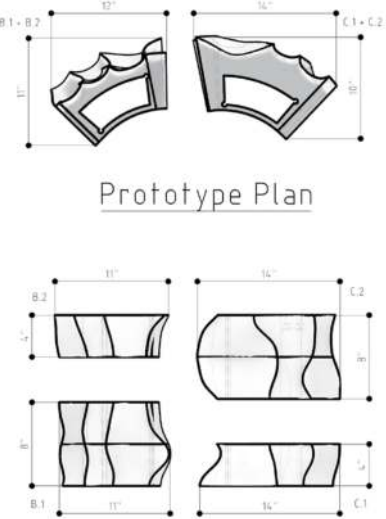
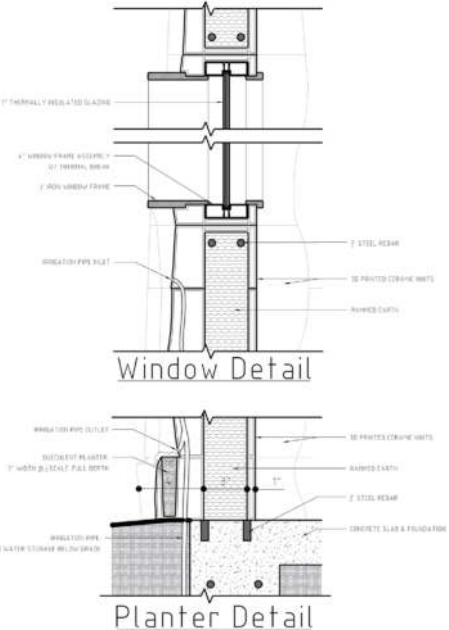
This project investigates the convergence of biological logic and digital fabrication, reimagining the architectural skin as a "living" mediator rather than a static barrier. By utilizing algorithmic growth patterns, Earthen explores how a building's facade can exhibit organic behaviors—expansion, contraction, and sensory feedback.

Wall Chunk Mockup @ 1/2 Scale**Ceramic Unit Schematics**

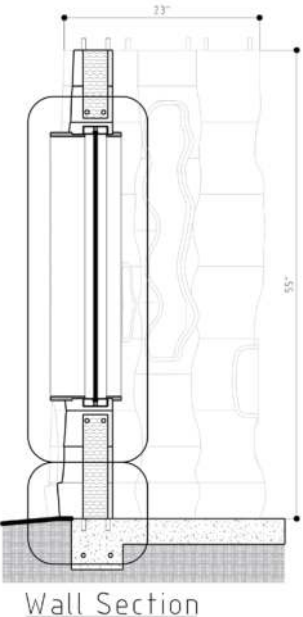
Chunk Prototype



Prototype Chunk
Mockup @ 1/2 Scale



Prototype Plan
Prototype Elevation



Glazing Pattern



Chrome/Palladium tone
thin band just inside the rim;
goes on first to melt and pull



Reactive white
covers entire piece



Colors on the edges
brushed over the chrome/
palladium band, slightly inside
the lip so they melt inward
and pool.





SCI-Arc Gala 2025

Instructor

M. Casey Rehm

Duration

2 weeks (Fall 2025)

3-person Teamwork

Developed image sequences, translated architectural projects into motion narratives, coordinated visual output for large-scale projection

How can architecture reach the public without being physically occupied?

A large-scale projection presented on the façade of the Los Angeles Central Library during the SCI-Arc Gala. The work brings architectural projects into a public setting, where they can be seen outside of their usual context.

People gather, pass by, and watch the projection from the street. The projection allows people in the city to encounter architectural work without entering a building or gallery. The projection turns the building façade into a temporary surface where architecture is shared with a broader audience.



This project changed how I think about representation. It showed that architecture can circulate through images and still carry spatial ideas, reaching people who may never enter the spaces themselves.

Using AI-generated imagery and animation, we developed a continuous sequence that connects student work across architecture, product design, and interiors. As each project transitions into the next, the surrounding environment shifts with it, allowing different spatial ideas to overlap within a single visual field.

